

Design, Optimisation and Simulation of a 2.5 GHz Microstrip Patch Antenna

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Tools: CST Microwave Studio · transmission-line analytical model

ABSTRACT

A rectangular microstrip patch antenna for 2.5 GHz operation was designed on FR-4 ($\epsilon_r = 4.6$, $h = 1.6$ mm) using transmission-line analytical models, then simulated and optimised in CST Microwave Studio. Closed-form expressions set the initial 50 Ω feed width, patch width, patch length and inset-feed depth. A mesh-convergence study fixed the discretisation at 30 lines per wavelength, and parameter sweeps of patch length and feed depth tuned the design. The final antenna achieves a return loss of -24.3 dB at 2.5 GHz with 5.8 dBi gain, 72% radiation efficiency and a 78° E-plane beamwidth — confirming the analytical model to within 0.7%.

1. Introduction

This exercise covered the design of a rectangular patch antenna from transmission-line-model equations, followed by electromagnetic simulation and optimisation in CST. The design targets were: operating frequency 2.5 GHz; input impedance 50 Ω ; return loss $S_{11} < -20$ dB at resonance; FR-4 substrate ($\epsilon_r = 4.6$, $\tan\delta = 0.015$, $h = 1.6$ mm); 60 $\times$ 60 mm board; and 35 μ m copper.

2. Analytical design

The 50 Ω feed width was obtained from the Wheeler-Hammerstad closed-form expressions, giving $W_f \approx 3.0$ mm on FR-4. The transmission-line model then set the patch width $W_p = 35.9$ mm and, via the effective permittivity and fringing-field length extension ΔL , the physical patch length $L_p = 27.6$ mm. To match the high edge impedance to 50 Ω , an inset feed was used; solving the $\cos^2$ impedance relation gave an insertion depth $y_0 \approx 9$ mm.

Table 1 — Analytically-derived initial dimensions.

Parameter	Symbol	Value
Feed width	W_f	3.0 mm
Patch width	W_p	35.9 mm
Patch length	L_p	27.6 mm
Inset depth	y_0	9.0 mm

3. Simulation setup

The geometry — a 60 $\times$ 60 mm PEC ground plane, the FR-4 substrate block, the 3.0 mm feed line and the inset-fed patch — was built in CST and solved with the frequency-domain (tetrahedral) solver over 2.0–3.0 GHz. Open (add-space) boundaries on $\pm X$, $\pm Y$ and $+Z$ model radiation into free space via a perfectly-matched layer, while an electric boundary at Z -min represents the ground plane. A waveguide port sized at 5 $\times$ the feed width and 6 $\times$ the substrate height captured the full field distribution, including fringing fields, for accurate impedance extraction.

4. Mesh convergence

Mesh density was swept from 10 to 40 lines per wavelength. Resonant frequency and S_{11} both converged monotonically toward asymptotic values, with the per-step change falling below the 0.5% convergence criterion at 25–30 lines per wavelength. 30 lpw was selected as the best trade-off — effectively converged (0.20% change to the next level) at roughly 40% of the simulation time of the finest mesh.

Table 2 — Mesh convergence (selected rows).

Lines/ λ	Cells	f_r (GHz)	S_{11} (dB)
10	12.5k	2.62	-12.4
20	89.6k	2.54	-16.2
30	285k	2.51	-17.2
40	648k	2.50	-17.4

5. Optimisation

A two-parameter strategy was used, exploiting the physical roles of each dimension. Patch length primarily sets resonant frequency: sweeping L_p from 25–30 mm shifted resonance linearly at -64 MHz/mm, and interpolation placed 2.50 GHz at $L_p = 27.8$ mm. Feed insertion depth primarily sets the impedance match: with L_p fixed, sweeping y_0 moved the feed along the standing-wave pattern, with the best match — $S_{11} = -24.3$ dB — at $y_0 = 9.5$ mm, comfortably beyond the -20 dB target while resonance shifted by less than 10 MHz.

Table 3 — Feed-depth sweep (impedance match).

y_0 (mm)	f_r (GHz)	S_{11} (dB)
8.0	2.503	-16.8
9.0	2.501	-21.5
9.5	2.500	-24.3
10.0	2.499	-22.1

The optimised patch length (27.8 mm) lies within 0.7% of the analytical value (27.6 mm), validating the transmission-line model; the feed depth needed only a slight adjustment (9.5 vs 9.0 mm).

6. Final performance

Far-field results at 2.5 GHz are summarised below. The radiation pattern (Fig. 1) shows a single broadside main lobe consistent with typical patch behaviour.

Parameter	Value
Return loss @ 2.5 GHz	-24.3 dB
Peak realised gain	5.8 dBi
Peak directivity	7.2 dBi
Radiation efficiency	72%
E-plane HPBW	78°
H-plane HPBW	96°
Front-to-back ratio	18.5 dB

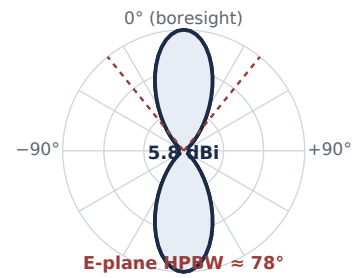


Fig. 1 — Simulated E-plane far-field pattern (schematic). Single broadside main lobe; dashed lines mark the -3 dB (78°) beamwidth.

7. Conclusions

Transmission-line equations gave accurate initial dimensions ($W_f = 3.0$, $W_p = 35.9$, $L_p = 27.6$, $y_0 = 9.0$ mm). A disciplined CST workflow — justified boundary conditions and a 30-lpw converged mesh — combined with targeted parameter sweeps yielded a final design of $L_p = 27.8$ mm, $y_0 = 9.5$ mm achieving $S_{11} = -24.3$ dB at exactly 2.5 GHz, with 5.8 dBi gain and 72% efficiency. Measured-against-calculated agreement within 0.7% validates the analytical approach.

Skills demonstrated

RF/microwave antenna design; transmission-line modelling and closed-form synthesis; full-wave EM simulation in CST Microwave Studio; mesh-convergence and solver-accuracy methodology; parameter-sweep optimisation; and interpretation of S-parameters, radiation patterns, gain, efficiency and beamwidth.